

Relationships between metabolic rate, muscle electromyograms and swim performance of adult chinook salmon

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Oxygen consumption rates of adult spring chinook salmon *Oncorhynchus tshawytscha* increased with swim speed and, depending on temperature and fish mass, ranged from $609 \text{ mg O}_2 \text{ h}^{-1}$ at 30 cm s^{-1} ($c. 0.5 \text{ BL s}^{-1}$) to $3347 \text{ mg O}_2 \text{ h}^{-1}$ at 170 cm s^{-1} ($c. 2.3 \text{ BL s}^{-1}$). Corrected for fish mass, these values ranged from 122 to $670 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$, and were similar to other *Oncorhynchus* species. At all temperatures (8, 12.5 and 17°C), maximum oxygen consumption values levelled off and slightly declined with increasing swim speed $>170 \text{ cm s}^{-1}$, and a third-order polynomial regression model fitted the data best. The upper critical swim speed (U_{crit}) of fish tested at two laboratories averaged 155 cm s^{-1} (2.1 BL s^{-1}), but U_{crit} of fish tested at the Pacific Northwest National Laboratory were significantly higher (mean 165 cm s^{-1}) than those from fish tested at the Columbia River Research Laboratory (mean 140 cm s^{-1}). Swim trials using fish that had electromyogram (EMG) transmitters implanted in them suggested that at a swim speed of $c. 135 \text{ cm s}^{-1}$, red muscle EMG pulse rates slowed and white muscle EMG pulse rates increased. Although there was significant variation between individual fish, this swim speed was $c. 80\%$ of the U_{crit} for the fish used in the EMG trials (mean U_{crit} 168.2 cm s^{-1}). Bioenergetic modelling of the upstream migration of adult chinook salmon should consider incorporating an anaerobic fraction of the energy budget when swim speeds are $\geq 80\%$ of the U_{crit} .

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Key words: Columbia River; electromyogram; hydroelectric dams; metabolic rate; spring chinook salmon; swim speed.

INTRODUCTION

Adult spring chinook salmon *Oncorhynchus tshawytscha* (Walbaum) migrate upstream to headwater spawning areas during periods of peak river discharges (Healey, 1991). Early-season migration increases the probability that adults will

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be able to access small tributary streams that may be inaccessible during the autumn when discharge is reduced. The construction of hydroelectric dams in the Columbia River basin, however, has altered the upstream migration corridor for anadromous salmonids (Ebel *et al.*, 1989) and now high river discharge in the basin potentially increases delay and fall-back (fish passing a dam and then falling back over the spillway) of adult spring chinook salmon at dams (Liscom *et al.*, 1985; Bjornn *et al.*, 1999; Reischel, 1999). Delay and fall-back is of concern because a lengthened migration period could deplete fixed energy reserves, resulting in reduced spawning success or increased pre-spawning mortality (Beiningen & Ebel, 1970; Gray, 1990; Berman & Quinn, 1991). If dam passage incurs an energetic cost to Columbia River spring chinook salmon that is excessive, their energy reserves could be depleted and they could die before spawning.

The energy used during upstream migration is affected by swimming patterns of individual fishes (Hinch *et al.*, 1996; Hinch & Rand, 1998; Standen *et al.*, 2002). A variety of swimming behaviours, including burst, sustained and prolonged swimming (Beamish, 1978), are used by fishes during long or arduous migrations to balance efficient energy use against the need to reach spawning areas in a timely manner (Hinch & Rand, 2000). Determining when various behaviours are used requires knowledge of the relationship between swimming performance, active metabolism and muscle electromyograms (EMG). The critical swimming speed (U_{crit}) is a measure of swimming performance, and is approximately the speed at which fishes fatigue in an incremental velocity test (Hammer, 1995). It is generally assumed that maximum oxygen uptake occurs at U_{crit} (Brett, 1982; Farrell & Steffensen, 1987) and this measure of performance is presumed to be a relatively close estimate of the maximum aerobic capacity of fishes (Hammer, 1995).

Studies of muscle EMG suggest that aerobic metabolism predominates in salmonids (*i.e.* red muscle activity) at swim speeds up to 70–80% of the U_{crit} , but anaerobiosis may contribute to salmonid energy budgets as swim speeds approach and exceed the U_{crit} (Hammer, 1995). Adult salmonids swim below and above U_{crit} during the upstream migration, using burst and coast strategies to move upstream (Hinch & Bratty, 2000; Hinch & Rand, 2000; Hinch *et al.*, 2002). Knowledge about when fishes utilize red and white muscle during upstream migration will provide information on the relative importance of aerobic and anaerobic metabolism to the allocation of an adult salmonid's energy budget. Although there have been studies conducted on these topics with sockeye salmon *Oncorhynchus nerka* (Walbaum) (Brett, 1964; Brett & Glass, 1973; Hinch *et al.*, 1996, 2002; Hinch & Bratty, 2000) and pink salmon *Oncorhynchus gorbuscha* (Walbaum) (Williams & Brett, 1987; Hinch *et al.*, 2002; Standen *et al.*, 2002), little information is available for adult chinook salmon.

A laboratory investigation was initiated in 2000 to accomplish the following objectives: 1) measure active metabolism of adult spring chinook salmon from the Columbia River at three water temperatures over a range of swim speeds; 2) estimate the U_{crit} of adult spring chinook salmon; 3) monitor EMG of red and white muscle in adult spring chinook salmon over a range of swim speeds to determine the relative importance of red and white muscle used in swimming.

MATERIALS AND METHODS

FISH COLLECTION AND HOLDING

A total of 65 adult spring chinook salmon were obtained in May and June 2000, from the Little White Salmon National Fish Hatchery, Cook, Washington, U.S.A. Fish were transported to either the Pacific Northwest National Laboratory (PNNL), Richland, Washington, or the Columbia River Research Laboratory (CRRL), Cook, Washington. Transport densities were kept low (<12 fish per trip), dissolved oxygen levels were maintained at $\geq 100\%$ air-saturation, and the water was treated with polyquaternary (160 ppm) and clove oil (c.10 ppm) to reduce stress. Water temperature during transport was maintained between 7 and 9°C which was similar to the hatchery. Following arrival, fish were randomly assigned to one of three tanks, and water temperatures in each tank were adjusted at a rate of 2°C per day (Cech, 1990) to a final temperature of 8, 12.5 or 17°C. Once the fish reached their test temperatures, they were acclimated for at least 1 week (range 7–12 days) before being used in experiments. Fish were treated every 3 or 4 days for 1 h with a static bath of formalin (37% formaldehyde) at a concentration of 0.17 ml l⁻¹ to prevent fungal infections (Piper *et al.*, 1982). Fish were not handled during the formalin baths, and treatments were administered at least 24 h prior to the fish being used in a trial. All morphometric measurements (fork length, L_F , mass and girth) were made with fish under sedation while being transferred from the holding tanks to the respirometers. Sex was determined by inspection of gonads post-mortem following completion of the study.

RESPIROMETRY TRIALS

Swimming respirometry was used to determine the relation between metabolic rate (*i.e.* oxygen consumption) and swim speed. A total of 45 adult spring chinook salmon were exercised at the two laboratories ($n = 24$ at PNNL and $n = 21$ at CRRL). Of the 45 fish used for the respirometry trials, 14 were males, 30 were females and one was not identified. Fork lengths ranged from 67.0 to 80.4 cm (mean $\pm$ s.d. 74.4 ± 3.1 cm) and masses ranged from 3.7 to 6.4 kg (5.0 ± 0.6 kg). Mean L_F and masses of fish were not significantly different between laboratories, sexes or temperatures (ANOVA, $P > 0.13$ for all factors).

Fifteen fish were tested at each of the three temperatures (*i.e.* 8, 12.5 or 17°C) using either a 5001 Blazka (CRRL) or a 22001 Brett-type (PNNL) respirometer. The working sections of the Blazka and Brett respirometers measured 142 and 180 cm long and 45.7 and 39.4 cm in diameter. Water velocities in both respirometers were created by a propeller driven by a variable-speed motor. Multiple measurements of velocity in the horizontal and vertical direction within the working section of the respirometers were made, and flow straighteners were adjusted on each device to achieve minimal variation in velocity profiles. Because of the smaller diameter of the Brett respirometer, a correction factor was used to adjust velocities for solid blocking effects (Smit *et al.*, 1971). During trials, both respirometers were sealed so that no gas exchange occurred. An electrified screen (Brett) or circular rings (Blazka) were used occasionally to keep fish from resting during trials.

Individual fish were anaesthetized with clove oil (50 ppm), weighed (to the nearest g), measured (L_F to the nearest mm) and placed in the respirometer. Fish were then acclimated overnight (>12 h) in the respirometer at a velocity of 30 cm s⁻¹ while being continually flushed with fresh water. Water from the respirometer was sampled (60 ml min⁻¹) using gas impermeable tubing and a peristaltic pump set in a continuous loop from the respirometer to a sealed Plexiglas chamber containing an oxygen electrode connected to an electronic meter (WTW Oxi 340 meter and CellOx 325 probe). The oxygen meter and probe were connected to a computer that logged data (time, per cent saturation of oxygen and dissolved oxygen concentration) every 5–15 s using software

supplied by WTW. The oxygen meter and probe were calibrated in air (*i.e.* 100% air-saturated) every day according to the manufacturer's instructions.

During the trials, freshwater inflow was turned off and the starting time, per cent saturation of oxygen and dissolved oxygen (mg l^{-1}) were recorded. Data were logged continuously until the dissolved oxygen levels had dropped $0.15\text{--}0.30\text{ mg l}^{-1}$ from the starting conditions. At the end of any speed increment, if the per cent air saturation was $>75\%$ and water temperature had not increased $>1^\circ\text{C}$ from the test temperature, the water velocity was increased to the next speed increment and data collection resumed. If these conditions were not met, the water velocity was reduced to 30 cm s^{-1} and the respirometer was flushed with fresh water until air-saturation was $>90\%$ and water temperature was reduced to starting conditions. Temperature increases $>1^\circ\text{C}$ during any given trial were rare (four fish with a maximum increase of 1.2°C), and most (*c.* 70%) of the trials were completed within 0.5°C of the starting conditions.

Using 20 cm s^{-1} increments, fish were tested at speeds (adjusted for blocking effects) ranging from 30 to 220 cm s^{-1} ; not every fish successfully completed all increments. Typically, at the lower velocities, fish were 'swum through' about three increments before flushing the respirometer. As velocities increased ($\geq 90\text{ cm s}^{-1}$), it was necessary to flush the respirometer with fresh water every one to two increments.

Data from fish at each velocity increment were plotted as the decrease in dissolved oxygen *v.* time and checked visually for a linear relation. If any increment was substantially non-linear or highly variable, the animal would be swum again at that speed, but this was rare. Following testing, fish were anaesthetized (50 mg l^{-1} MS222 or *c.* 10 ppm clove oil), removed from the respirometers, uniquely marked with a coloured Floy tag, and returned to their holding tank. A blank trial (30 min) was performed after each fish to determine the amount of microbial oxygen use. Microbial oxygen use was normally undetectable but when present was subtracted from the oxygen use estimate.

The oxygen consumption rate (MO_2 in $\text{mg O}_2\text{ h}^{-1}$) for an individual fish during a velocity increment was determined using the following equation: $MO_2 = A \cdot V$, where V was the volume of the respirometer (l) and A was the rate of oxygen consumed ($\text{mg O}_2\text{ l}^{-1}\text{ h}^{-1}$), minus the blank, during a velocity increment as determined by linear regression analysis of the reduction in dissolved oxygen levels over time (t): $A = b_0 + (t \cdot b_1)$.

A series of polynomial linear regression models were compared to determine the best model for predicting oxygen consumption as a function of swim speed for each fish. Because each fish was tested at multiple swim speeds but at only a single temperature, the resulting polynomial coefficients for each fish provide an independent estimate of how an individual fish with a given mass would respond at a given temperature across a series of swim speeds. Polynomial regression was used to remove patterns of curvature from the residuals. For all analyses, the $\ln MO_2$ were used to reduce the residual variance and within-class heterogeneity.

ANOVA on the estimated slopes and intercepts for each fish was then used to examine the differences across temperatures, laboratories, sex and fish mass (as a covariate) to determine the level of complexity in a regression model combining fish. Regression of the estimated slopes and intercepts on body size and temperature was used to determine the combined model for all fish (Minitab, 2000).

CRITICAL SWIM SPEED TRIALS

A total of 22 adult spring chinook salmon were used to estimate U_{crit} ($n = 13$ at PNNL and $n = 9$ at CRRL). Of the 22 fish used for the swimming performance tests, seven were males, 13 were females and two were not identified. Fork lengths ranged from 69 to 82 cm (mean $\pm$ s.d. $75.5 \pm 3.7\text{ cm}$) and masses ranged from 3.6 to 6.4 kg ($5.1 \pm 0.6\text{ kg}$). Mean masses of fish were not significantly different between laboratories ($P = 0.41$), but L_F of fish tested at CRRL were significantly greater (ANOVA, d.f. = 2 and 20, $P = 0.02$) than fish tested at PNNL (mean 77.7 *v.* 73.9 cm). Fish used in these trials were either some of those used in the respirometry trials or were unused fish. In all cases, fish were

allowed at least 12 to 18 h of recovery after a respirometry trial before being used in a U_{crit} trial. Water temperature for all U_{crit} trials was 12.5°C.

Fish were anaesthetized, placed in a respirometer (same as those used in respirometry experiments) and allowed at least 1 h of recovery under flow-through conditions (water velocity 30–50 cm s⁻¹). After recovery, the water velocity was increased to 75 cm s⁻¹ and the fish was swum for 30 min. Water velocity was then increased 25 cm s⁻¹ every 30 min until the fish fatigued. Fatigue was defined as when the fish rested or became impinged on the downstream screen and would not leave despite several electric shocks. Critical swim speed was calculated using the formula described by Beamish (1978). For fish swum in the Brett respirometer, velocities were corrected for solid blocking effects (Smit *et al.*, 1971), as described previously. U_{crit} values were compared between laboratories, sex and fish size by ANOVA.

ELECTROMYOGRAM-SWIM SPEED TRIALS

To document the relation between EMG activity and swimming performance, EMG radio transmitters were surgically implanted into 10 fish that were then swum at defined speeds in the Brett respirometer at PNNL. The fish used in these experiments were a subset of those used in the U_{crit} tests and the EMG studies were completed following the U_{crit} experiments. Fish were allowed 48 h for recovery prior to starting the EMG swim trials. Of the 10 fish used for the EMG studies, three were males and seven were females. Fork lengths ranged from 69 to 80 cm (mean $\pm$ s.d. 73.9 $\pm$ 3.7 cm) and masses ranged from 3.6 to 6.4 kg (5.0 $\pm$ 0.8 kg); L_F and masses did not differ significantly between sexes (ANOVA, $P > 0.05$).

Electromyogram transmitters were implanted using methods similar to Hinch *et al.* (1996). Two transmitters were placed in each fish. Electrodes were positioned along the lateral line at $c. 0.4 L_F$. For placement in red muscle, the electrode anchors were inserted just under the surface of the skin. For placement in white muscle, the electrode anchors were inserted $c. 1$ cm below the surface of the skin. The electrical impulses emitted by the muscles were detected by the EMG tag and when a factory-set threshold (in μ V) was reached a radio pulse was transmitted. Therefore, increasing muscle activity resulted in an increase in the pulse rate of transmitted radio signals (*i.e.* the time between the radio transmissions decreased; Kaseloo *et al.*, 1992).

Using a telemetry receiver, 30 EMG values were recorded from fish completely anaesthetized (50 ppm clove oil). These values were assumed to be representative of fish at rest. Following 1 h of recovery in the respirometer under flow-through conditions and velocity set at 30 cm s⁻¹, EMGs were recorded while fish were swum at speeds ranging from 30 to 220 cm s⁻¹ at 20 cm s⁻¹ increments. Fish would be encouraged to swim by briefly electrifying the downstream screen. Fish swam at each speed until at least 30 data points were recorded (generally only a few minutes) before the velocity was increased to the next speed.

After the last speed increment, the velocity was reduced to 30 cm s⁻¹, and data were examined by plotting mean EMG output at each swim speed. Fish were re-tested at any speed that was an obvious outlier on the observed trend line, but this was rare. When a trial was complete, the fish was removed from the respirometer, placed in an anaesthetic bath (50 ppm clove oil) and sacrificed. The electrodes were dissected out to record the distance from the nose to the centre of the electrodes, the distance between electrodes, the depth of the electrodes below the skin and electrode orientation and location (red or white muscle).

Data analysis was accomplished using linear regression which fitted muscle EMG data to swim speed. The base model took a piece-wise linear form on two model segments: $EMG(x) = a_1 + b_1x$ ($x < x_0$) and $EMG(x) = a_2 + b_2(x - x_0)$ $x \geq x_0$, where $x_0 = (a_2 - a_1) b_1^{-1}$. This was a generalization of a model form that has been used in toxicological threshold models and is sometimes referred to as the 'hockey stick model' (Klepper & Bedaux, 1997). These models were fitted to red and white EMG muscle data separately. The fitting procedure estimates four model coefficients (intercepts: a_1 , a_2 ; slopes: b_1 , b_2) with the breakpoint between model segments (x_0) expressed as a function of a_1 , a_2

and b_1 . The model is thus non-linear in x . The fitting procedure uses an iteratively re-weighted least squares (IRLS) methodology with a Gauss-Newton optimization algorithm (SAS, 1999). In this manner, both model segments are fit simultaneously and meet at the breakpoint (x_0).

RESULTS

RESPIRATORY METABOLISM

At all three temperatures, oxygen consumption increased with increasing swim speed (Fig. 1). The lowest oxygen consumption occurred at the lowest swim speed (30 cm s^{-1}) and increased with increasing temperature (Table I). The highest oxygen consumptions occurred at average swim speeds approaching 170 cm s^{-1} , and were highest at the intermediate temperature (Table I). Maximum oxygen consumption was 4.5–4.7 times higher than the minimum oxygen consumption at all three temperatures.

A third-order polynomial linear regression model of $\ln$ oxygen consumption *v.* swim speed was fitted to each fish so that the third-order slope was common for all fish, but intercepts and first and second order slopes differed (Fig. 1). Analysis of covariance (ANCOVA) of the 45 individual third order polynomial model variables for each laboratory, sex, temperature class, laboratory–temperature interaction and $\ln$ transformed mass (the covariate) indicated that these factors had little influence on the first and second order slope coefficients associated with swim speed ($P > 0.13$). Fish mass, however, was a significant covariate of the intercept ($P = 0.001$; Table II) on the MO_2 -swim speed relationships. None of the laboratory–temperature interactions were significant ($P > 0.30$). There was also no statistical difference in the intercepts of each individual fish based on water temperature class ($P > 0.06$), although at the lowest swim speed there was a strong trend of the oxygen consumption with increased water temperature (Table I). The first- and second-order polynomial slopes were not significantly different between temperatures ($P > 0.46$) and did not display a trend.

Regression lines of the minimum and maximum MO_2 ($\text{mg O}_2 \text{ h}^{-1}$) *v.* body mass (kg) were fitted by least squares of the $\ln$ - $\ln$ transformed data (Fig. 2). Equations for the regression results for each data series are presented in Table III. With the exception of the lowest swim speed at 8°C , slope (b) values were significant and ranged from 1.10 to 1.79.

A regression model of $\ln$ transformed mass and temperature on the individual fish intercepts was used to estimate the base-line oxygen consumption component of the combined fish model. The remaining model components associated with swim speed were estimated as the average slope coefficients associated with the swim speed. All of the slopes were significantly greater than zero (Table IV). This model explained 93% of the total variance in $\ln MO_2$ ($\text{mg O}_2 \text{ h}^{-1}$). Although the second- and third-order slope coefficients were extremely small ($< 2 \times 10^{-4}$), these coefficients were retained in the model because they reflect the ‘flattening out’ of oxygen consumption observed at the higher swim speeds. Due to round-off error, all decimal places should be used.

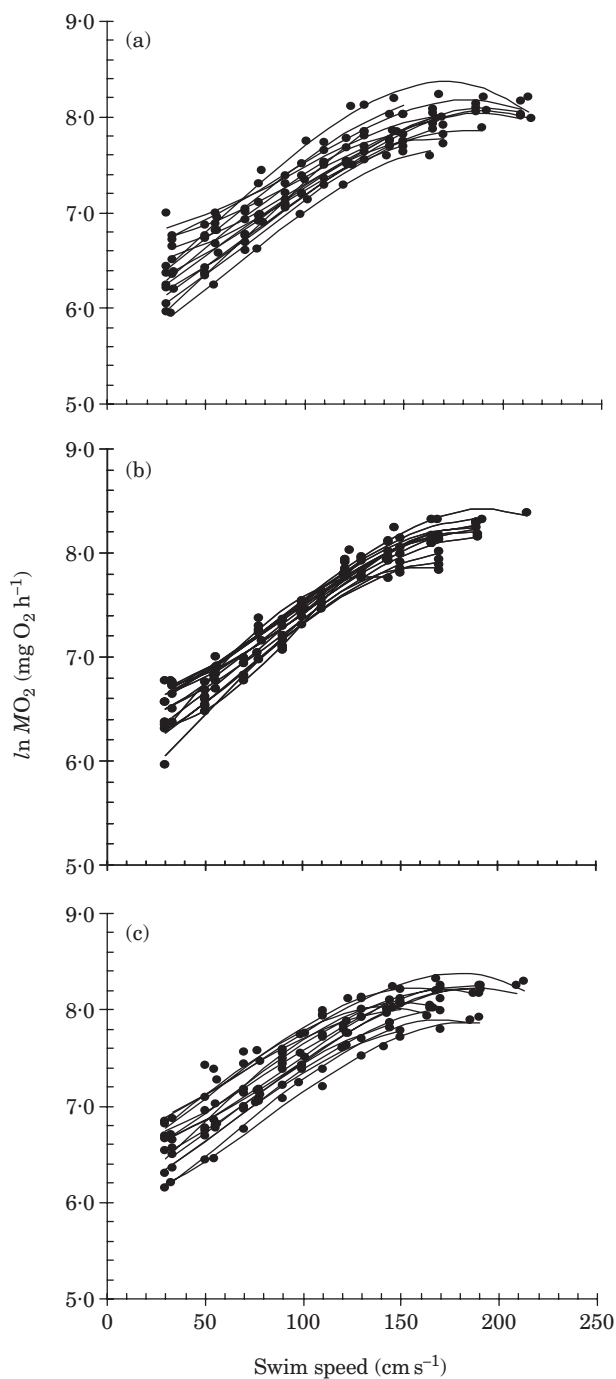


FIG. 1. The observed oxygen consumption (●) and fitted third order polynomial models for each of 45 adult spring chinook salmon as a function of swim speed at (a) 8, (b) 12.5 and (c) 17°C.

TABLE I. The minimum and maximum oxygen consumption rate of adult spring chinook salmon at three temperatures. Sample size was 15 fish at each temperature. The average masses of fish tested at 8, 12.5 and 17°C were 5.1, 5.1 and 4.8 kg, respectively

Temperature (°C)	Average swim speed (cm s ⁻¹)	Mean ± s.e. $\dot{M}O_2$ (mg O ₂ h ⁻¹)
Minimum		
8	30	609.2 ± 45.3
12.5	30	708.6 ± 38.0
17	30	720.2 ± 38.2
Maximum		
8	164.5	2869.8 ± 137.6
12.5	167.1	3346.5 ± 160.4
17	168.0	3268.2 ± 134.6

CRITICAL SWIM SPEED

The U_{crit} ranged from 125 to 196 cm s⁻¹ (mean ± s.d. 154.7 ± 21.4 cm s⁻¹). The U_{crit} values did not differ by sex (ANOVA, d.f. = 1 and 18, $P = 0.5874$) or size (d.f. = 1 and 20, $P = 0.75$). The CRRL fish had significantly lower measured U_{crit} (ANOVA, d.f. = 1 and 20, $P = 0.005$) with a mean ± s.d. of 140.2 ± 16.9 cm s⁻¹ v. 164.8 ± 18.6 cm s⁻¹ at PNNL.

ELECTROMYOGRAM-SWIM SPEED RELATIONS

The U_{crit} of the 10 fish used in the EMG-swim speed trials ranged from 145 to 196 cm s⁻¹ (mean ± s.d. 168.2 ± 18.8 cm s⁻¹). Red muscle EMG signals (pulse min⁻¹) increased with increasing swim speed for all fish but there were individual differences in the rate of increase (Fig. 3). For example, some fish showed a plateau in red muscle EMG pulse rates at higher swim speeds (fish 1, 2, 3, 5, 6 and 9) while others did not appear to slow down even at swim speeds that exceeded U_{crit} (fish 4, 7, 8 and 10). Individual fish also had different white muscle EMG signals with some fish demonstrating increasing recruitment at moderate swim speeds (fish 1, 6 and 7) while others showed no change until fast swim speeds (fish 3, 4, 5, 8 and 9). There was little difference in red and white muscle EMG patterns in fish 10 with red and white muscle EMGs increasing

TABLE II. Regression results from ANCOVA on intercepts

Source	d.f.	Sequential sum of squares	Adjusted sum of squares	Adjusted mean square	<i>F</i>	<i>P</i>
Ln mass	1	1.08699	1.29282	1.29282	13.13	0.001
Laboratory	1	0.29665	0.26143	0.26143	2.65	0.112
Temperature class	2	0.53523	0.5945	0.29725	3.02	0.061
Laboratory × temperature class	2	0.24397	0.24397	0.12198	1.24	0.301
Error	38	3.74287	3.74287	0.0985		
Total	44	5.9057				

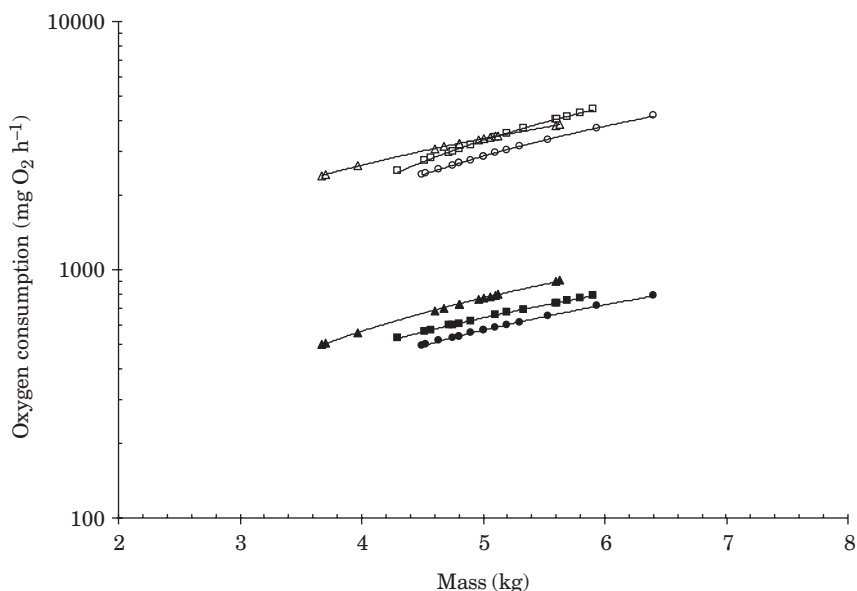


FIG. 2. The relationship between metabolic rate and mass for adult spring chinook salmon tested at 8 (○, ●), 12.5 (□, ■) and 17°C (△, ▲) described by the power function $y = ax^b$ (slope and intercept coefficients are shown in Table III). (●, ■ and ▲ minimum values; ○, □ and △ maximum values).

throughout the entire range of swim speeds. Finally, white muscle EMG pulse rates initially decreased in fish 2 only to increase as swim speeds increased.

Although the EMG signals at each swim speed varied between individuals, the breakpoints at which the red and white muscle EMG pulse rates converged are nearly synchronized in their point estimates and confidence bounds in the general model. This convergence suggested a nearly simultaneous shift from increasing red muscle activity ($<135 \text{ cm s}^{-1}$) to increasing white muscle activity ($>137 \text{ cm s}^{-1}$; Fig. 4). In other words, although red muscle EMG signals did not completely stop, their rate of increase slowed at about the same time that the EMG pulse rate increased in white muscle. Goodness-of-fit tests were highly

TABLE III. Coefficient estimates $\pm$ S.E. of the regression of $\ln MO_2$ ($\text{mg O}_2 \text{ h}^{-1}$) and $\ln$ mass (kg). Minimum and maximum metabolic activity occurred at swim speeds of 30 and 170 cm s^{-1} , respectively

Temperature ($^{\circ}\text{C}$)	a (intercept)	$P a$	b (slope)	$P b$
Minimum				
8	4.25 ± 1.20	0.004	1.30 ± 0.74	0.101
12.5	4.48 ± 0.72	<0.0001	1.23 ± 0.44	0.015
17	4.42 ± 0.52	<0.0001	1.38 ± 0.33	0.001
Maximum				
8	5.48 ± 0.60	<0.0001	1.54 ± 0.36	0.001
12.5	5.22 ± 0.95	<0.0001	1.79 ± 0.58	0.009
17	6.35 ± 0.40	<0.0001	1.10 ± 0.26	0.001

TABLE IV. Regression results from the combined-fish third order polynomial model ($n=45$ fish, d.f. = 2 and 44). Speed refers to the swim speed of the fish with speed² and speed³ the second and third order coefficients for this variable

Variable	Coefficient	<i>t</i>	<i>P</i>
Intercept	3.1800 ± 0.7602	4.18	0.0001
Ln mass (kg)	1.6400 ± 0.4366	3.76	0.0010
Temperature (°C)	0.0315 ± 0.0133	2.36	0.0230
Speed (cm s ⁻¹)	0.00601 ± 0.00090	6.409588	0.0000
Speed ² (cm s ⁻¹)	$0.00011 \pm 4.5 \text{ E} - 06$	24.59675	0.0000
Speed ³ (cm s ⁻¹)	-0.00000048 ± 0		

significant (ANOVA, $P < 0.0001$) for both regression models (both models d.f. regression = 4, d.f. residual-red = 2573, d.f. residual-white = 2569; Table V). This convergence point (135–137 cm s⁻¹) is *c.* 80% of the mean U_{crit} for these fish (95% CI, 73–88%). For all fish, post-mortem dissections showed that all electrodes were firmly implanted in the appropriate muscle type. No malfunctioning transmitters during this phase of the study were noted, but fish were not re-swum to determine if transmitter drift occurred over time.

DISCUSSION

Oxygen consumption rates of adult spring chinook salmon increased with swim speed, and depending on temperature and corrected for fish mass, ranged from a minimum of 122 mg O₂ kg⁻¹ h⁻¹ at 30 cm s⁻¹ (*c.* 0.5 BL s⁻¹) to a maximum of 670 mg O₂ kg⁻¹ h⁻¹ at 170 cm s⁻¹ (*c.* 2.3 BL s⁻¹). The results suggest that oxygen consumption rates of adult spring chinook salmon are consistent with other Pacific salmonid adults tested at similar temperatures and swim speeds. For example, adult fall chinook salmon (4.6–14.0 kg) from the Columbia River tested over a range of swim speeds (0.5–1.4 BL s⁻¹) at 15 and 20°C consumed 150 to *c.* 400 mg O₂ kg⁻¹ h⁻¹ (Geist *et al.*, 2000). Oxygen consumption rates for adult sockeye salmon (1.5–2.2 kg) tested at 8–10°C ranged from 150 to 513 mg O₂ kg⁻¹ h⁻¹ (Hinch *et al.*, 1996) while adult sockeye salmon (1.4 kg) tested at 15°C ranged from a minimum observed rate of 150 mg O₂ kg⁻¹ h⁻¹ to a maximum level of 717 mg O₂ kg⁻¹ h⁻¹ (Brett, 1965). The highest levels of oxygen consumption for adult Pacific salmonids were measured on pink salmon (1.6 kg) tested at 15°C and ranged from *c.* 150–830 mg O₂ kg⁻¹ h⁻¹ (Milliken, 1983; Williams *et al.*, 1986), although Brett (1995) noted that these values may be artificially elevated due to excitability of fish while they were being tested.

The results confirm that like other adult salmonids, spring chinook salmon have a wide scope of activity (Fry, 1957). The ratio of maximum to minimum levels of oxygen consumption for adult spring chinook salmon (range 4.5–5.0) was consistent with other species of salmonids. Although Brett (1965) reported a 16-fold increase in metabolic rate of sockeye salmon from the extrapolated standard rate to the full active rate, this ratio is closer to 5 if the actual oxygen consumption values measured at swim speeds of 0.5 and 2.0 BL s⁻¹ are

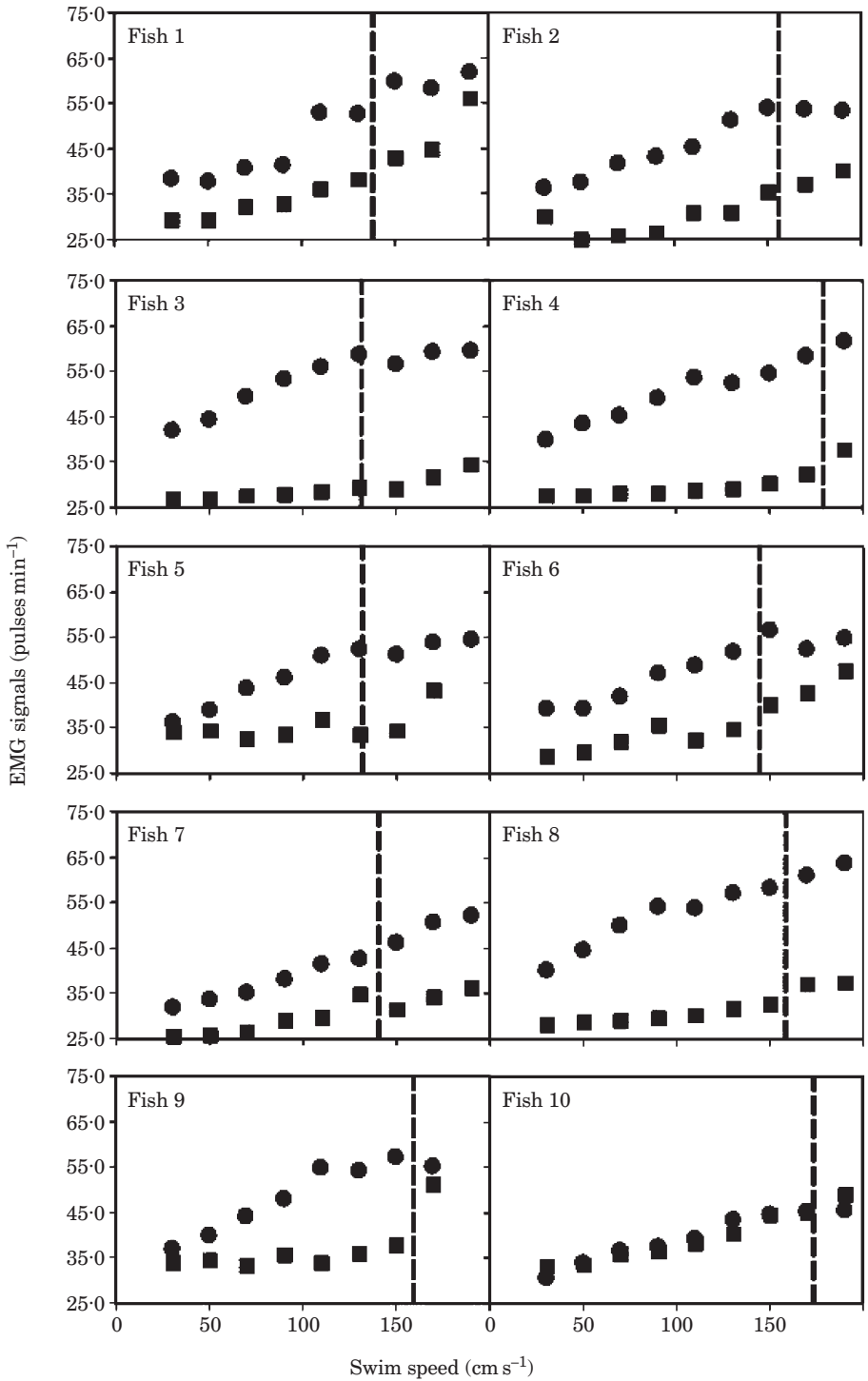


FIG. 3. Red (●) and white (■) muscle electromyograms and swim speed for 10 adult spring chinook salmon. ---, individual-specific U_{crit} .

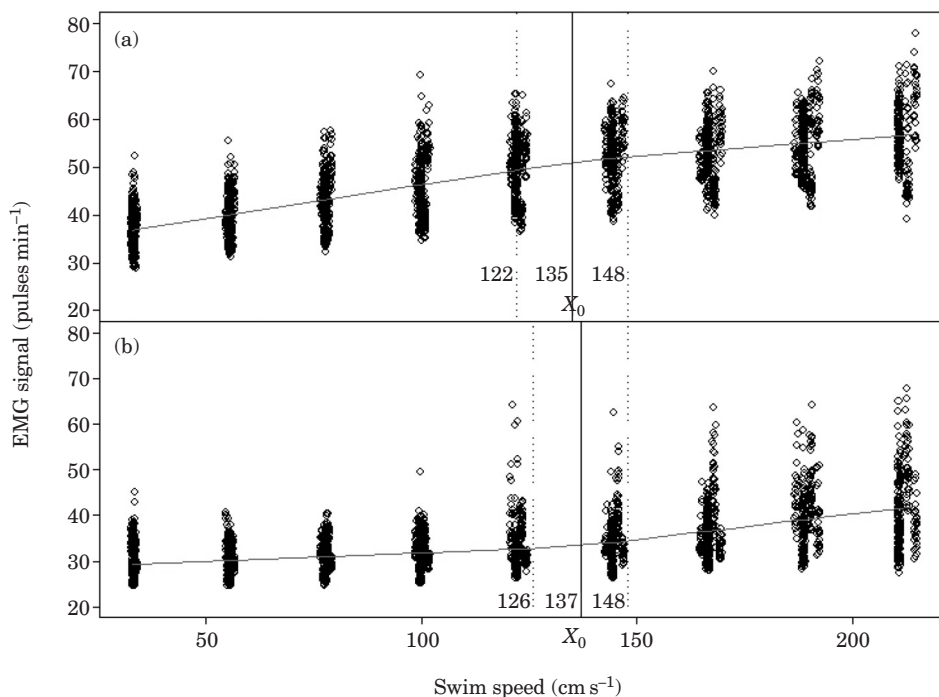


FIG. 4. Fitted models for (a) red and (b) white muscle electromyograms and swim speed for 10 adult spring chinook salmon. An approximate 95% CI is plotted for the estimated breakpoint (X_0) between model segments.

compared. Adult pink salmon (1.1–2.5 kg) swum at 15°C also showed a five-fold increase in minimum to maximum oxygen consumption (Milliken, 1983; Williams *et al.*, 1986) and Rao (1968) reported a five-fold increase in oxygen consumption between the lowest and highest swim speed for rainbow trout *Oncorhynchus mykiss* (Walbaum) tested at 15°C. A wide scope of activity is beneficial to long-distance migrants that are trying to efficiently use fixed energy reserves because migration through complex hydraulic environments requires

TABLE V. Regression results for the two linear regression models relating red and white muscle EMGs to swim speed

Source	d.f.	Sum of squares	Mean square	<i>F</i>	<i>P</i>
Red EMG					
Regression	4	5 968 213	1 492 053	44 580	<0.0001
Residual	2573	86 116.6	33.5	—	—
Total	2577	6 054 329.6	—	—	—
White EMG					
Regression	4	2 961 919	740 480	24 761	<0.0001
Residual	2569	76 825.3	29.90	—	—
Total	2573	3 038 744.3	—	—	—

the ability to move from stationary positions to maximum aerobic metabolism and beyond over short distances. For example, 'burst and coast' swimming is a strategy of upstream locomotion (Hinch & Bratty, 2000; Hinch *et al.*, 2002) that requires fishes to utilize their full metabolic range. 'Burst and coast' strategies that take advantage of micro-scale features of river channels are efficient uses of energy because they exceed upper U_{crit} for only very short periods without resulting in fatigue (Hinch & Bratty, 2000). Therefore, strong selection pressures have been placed on the ability of adult salmonids to rapidly increase their metabolic rate from relatively low routine or resting rates to maximal rates where both aerobic and anaerobic metabolism is occurring (Leonard *et al.*, 1999).

There was a significant correlation between mass and metabolic rate for adult spring chinook salmon. The mass exponent b -value (*i.e.* the rate of change of metabolic rate with size) determined from the regression of $\ln MO_2$ ($\text{mg O}_2 \text{ h}^{-1}$) and $\ln$ mass exceeded a value of 1.0 and suggested that MO_2 per unit mass increased with increasing size at the minimum and maximum level of activity for all three temperatures (Fig. 5). Studies reporting b -values that are >1 are uncommon, particularly in anadromous salmonids (Brett, 1995) and, until this study, not available for adult salmonids >5 kg. The notable exception is with sockeye salmon (Brett, 1964, 1965; Brett & Glass, 1973). No relation between mass and fish size was found in young sockeye salmon (mean masses 33–63 g) at the standard or active metabolic rates, which was attributed to the narrow range of fish sizes tested (Brett, 1964; Brett & Glass, 1973). Subsequent studies using a wider range of fish sizes (mean mass 3–1400 g) did show a relation between size and metabolic rate of sockeye salmon at 15°C with $b=0.846$ at 25% of the maximum sustained swim speed and $b=0.970$ at the maximum speed ($2.65\text{--}6.65 \text{ BL s}^{-1}$, depending on fish size; Brett, 1965). Values of mass-exponents during active swimming have also been reported for brook trout *Salvelinus fontinalis* (Mitchill) (0.94 at 5°C and 0.75 at 20°C; Job, 1955) and rainbow trout (0.62–1.19 depending on temperature and salinity; Rao, 1968). The relation between metabolic rate and fish size should be established using fishes that vary in size over several orders of magnitude (Brett, 1964; Cech, 1990). Therefore, although the b -values reported here for adult spring chinook salmon are quite high, a wider range of sizes should be studied further before these mass exponents be applied to other size ranges within this species.

Although the differences between the intercepts and linear slopes of the models at 8, 12 and 17.5°C were minor, there was a strong trend of increased oxygen consumption at the lowest swim speed as water temperature increased. This is consistent with other studies of Pacific salmonids where standard or routine rates of oxygen consumption increase with increasing temperature (Brett, 1964, 1967; Beamish, 1964; Brett & Glass, 1973). Maximum active metabolic rates, however, were highest at 12.5°C, lowest at 8°C, and intermediate at 17°C. The Q_{10} values ranged from 0.95 (12.5–17°C) to 1.38 (8–12.5°C). Brett (1964) also found that active metabolic rates were highest at intermediate temperatures in juvenile sockeye salmon and attributed this to oxygen consumption rates being curtailed above an optimum temperature of 15°C due to a reduction in oxygen availability at higher temperatures. Slight reductions in dissolved oxygen concentration below air-saturation resulted in a

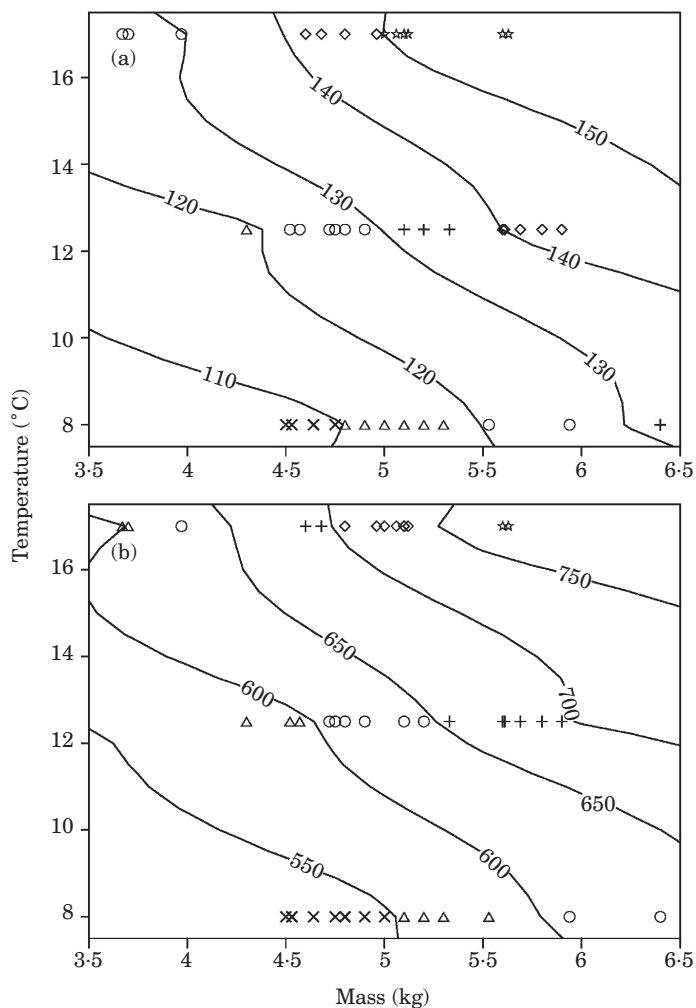


FIG. 5. Temperature-mass response surfaces for (a) minimum (swim speed 30 cm s^{-1}) and (b) maximum (swim speed 170 cm s^{-1}) metabolic activity ($\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$) of adult spring chinook salmon. (a) 100–110 ($\times$), 110–120 (Δ), 120–130 ($\circ$), 130–140 ($+$), 140–150 ($\diamond$) and 150–162 ($*$) $\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$. (b) 500–550 ($\times$), 550–600 (Δ), 600–650 ($\circ$), 650–700 ($+$), 700–750 ($\diamond$) and 750–800 ($*$) $\text{mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$.

reduction in maximum swimming performance in coho *Oncorhynchus kisutch* (Walbaum) and chinook salmon (Davis *et al.*, 1963). Management of the Columbia River for hydroelectric development has resulted in warming of the river in late spring and early summer over the last five decades so that it reaches 15.5°C *c.* 15 days earlier now than in 1954 (Quinn & Adams, 1996; Quinn *et al.*, 1997). The results reported here suggest reduced maximum metabolic performances of spring chinook salmon at water temperatures $>12.5^\circ \text{C}$. Therefore, even though water velocities are reduced in reservoir environments, which has resulted in upstream migration rates (*i.e.* distance per time) increasing since hydropower development (Bjornn & Peery, 1992), there is a tradeoff because metabolic demand increases with increasing temperatures. Consequently, delay

or fall-back at hydropower dams could reduce available energy reserves (Geist *et al.*, 2000).

The upper U_{crit} of fish tested at both laboratories averaged 155 cm s^{-1} (2.1 BL s^{-1}), but U_{crit} at PNNL were significantly higher (mean 165 cm s^{-1}) than those at CRRL (mean 140 cm s^{-1}). Brett (1965) reported critical swimming speeds for adult sockeye salmon of 2.5 BL s^{-1} , while Williams & Brett (1987) presented U_{crit} of adult pink salmon that ranged from 2.2 to 2.4 BL s^{-1} . These results suggest adult spring chinook salmon are relatively strong swimmers. Comparisons of U_{crit} between species and studies, however, is complicated by differences in study protocol (*e.g.* duration of tests and fixed *v.* incremental velocities), experimental conditions (temperature and dissolved oxygen) and other factors (sex and stage of maturation). For example, although there was no difference in U_{crit} between male and female spring chinook salmon, Williams & Brett (1987) reported that adult male pink salmon swam slightly better than female pink salmon. These contradictions are probably due to chinook salmon in this study not exhibiting distinct secondary sexual development, whereas adult pink salmon develop humped backs and hooked noses which probably affect their hydrodynamics (Brett, 1995). Differences in factors such as training, rearing condition, muscle mass and muscle enzyme profiles could also contribute to differences in U_{crit} values (Webb, 1995). These factors could have contributed to differences in U_{crit} values between the two laboratories as well. Although fish were swum in a Brett-style respirometer at PNNL and a Blazka-style respirometer at CRRL, differences due to respirometer type were assumed to be minor because both respirometers were calibrated using similar methods and exhibited uniform velocities. Fish at CRRL were slightly larger in length (78 *v.* $74 \text{ cm } L_F$) but these differences are not expected to have influenced swimming performance and do not conform to previous studies that show absolute swimming speeds increase with fish length (Bainbridge, 1958; Brett, 1965). Thus, differences between laboratories and inherent difficulties in comparing U_{crit} between studies suggests caution be used when comparing the swimming ability of adult spring chinook salmon to other fishes.

In spring chinook salmon, EMG activity in the red muscle was strongly and positively correlated with swim speed, a finding consistent with other studies on different species (McKinley & Power, 1992; Booth *et al.*, 1997; Thorstad *et al.*, 2000; Geist *et al.*, 2002). The majority (60%) of the fish studied showed a plateau in red muscle EMG values at swim speeds approaching or exceeding their U_{crit} , while in some fish (40%) the red muscle EMG pulse rates were increasing even at the highest swim speeds tested. White muscle EMG values also showed significant individual variation with two patterns observed. In the first pattern (*e.g.* fish 1, Fig. 3), white muscle EMG values slowly increased at relatively low swim speeds before increasing at a faster rate throughout the range of speeds tested. In the second pattern (*e.g.* fish 3, Fig. 3), there was little or no change in white muscle EMG values until swim speeds approached the U_{crit} values.

Precise positioning of electrodes within the axial musculature of the fish affected the range of biotelemetry signals emitted during activity of Atlantic salmon *Salmo salar* L. (Beddow & McKinley, 1999). Post-mortem dissections of EMG transmitters revealed no remarkable differences in electrode placement in

red and white muscle in spring chinook salmon, although placement within individual myomeres was not evaluated. The EMG tags used in this study incorporated an electrode anchor that measured *c.* 1 cm in length, and it is likely that EMG activity was recorded from multiple myomeres. The anchors are necessary to ensure that the electrodes stay in place during a long-term study, however, they may not be suitable for detailed investigations of EMG activity in specific muscle types. Beddow & McKinley (1999) noted smaller than expected signals of muscle activity at swim speeds $>U_{crit}$ when using an EMG transmitter as compared to external electromyography. They attributed this difference to the fact that the bandwidth filter on the EMG transmitter removed some of the higher frequency white muscle EMG signals, resulting in more white muscle EMG activity necessary to elicit a response. Although not the objective of this study, fine-scale micro-surgery using conventional electromyography may be required to assess variation between individual myomeres of various muscle types.

Although electrode placement may have contributed to individual variation in EMG values, it is probable that most of the variation was due to individual fish performance. Individual variation in swimming performance can be substantial in anadromous fishes (Hinch & Bratty, 2000; Hinch *et al.*, 2002). Leonard *et al.* (1999) observed different individual American shad *Alosa sapidissima* (Wilson) of the same sex, size and condition swimming at markedly different levels of performance. The authors suggested that individual variation in American shad may be due to simple differences in individual motivation during different periods of the upstream run, and believe this variation is important to their long-term survival (Leonard *et al.*, 1999). The results from this study indicate that within a population of chinook salmon, there is a mix of swim strategies, and, as suggested by Hinch & Rand (2000), such individual variation may confer an evolutionary benefit.

Evidence suggests that although aerobic metabolism dominates in salmonids during sustained and prolonged swimming, anaerobiosis may contribute to salmonid energy budgets at speeds $<U_{crit}$ (Jones, 1982). Muscle physiology and electromyography studies of muscle activity during sub-maximal swimming indicate that 80% of the U_{crit} may be the threshold for when white muscle is recruited into swimming (Hudson, 1973; Bone *et al.*, 1978; Jones, 1982; Hammer, 1995). Similar results were found with chinook salmon where white muscle EMG rates increased at *c.* 80% of the U_{crit} (95% CI 73–88% U_{crit}). Hinch *et al.* (2002) showed that adult male sockeye salmon spent $>50\%$ of their time bursting during passage through a constricted reach on the Fraser River, and suggested that salmonid energy budgets consider the importance of anaerobic metabolism. Burgetz *et al.* (1998) expressed the contribution of anaerobic metabolism during swimming in rainbow trout as a percentage of the rate of aerobic oxygen consumption at a given speed. That is, they applied an anaerobic tax which was based in large part on the metabolism of lactic acid produced during swimming. As swim speeds increased from 70 to 100% of the U_{crit} , the anaerobic tax increased such that energy used at 70, 80 and 100% of the U_{crit} would be an additional 65, 69.2 and 205.0% of the aerobic fraction of energy use. Whether such an analysis can be applied to other fishes is unknown, but these results suggest that estimates of the total energy used by adult spring chinook salmon during their upstream migration may be more accurate if they

incorporate the contribution of anaerobic metabolism at swim speeds $>70\text{--}80\%$ of the U_{crit} .

Efficient use of stored energy during their upstream spawning migration is essential to the successful reproduction of anadromous salmonids. Energetic costs of transport can be minimized, and reproductive fitness maximized, through use of different swimming strategies that are at or above U_{crit} (Webb, 1995; Hinch & Rand, 2000). The total energy used by adult salmonids during their upstream migration has important management implications. The physiological limitations of anadromous species during swimming should be considered as they move through significantly altered watersheds (Rand & Hinch, 1998). Upstream migration may impose significant costs to the available energy reserves of even strong swimmers such as salmonids, which may contribute to a reduction in reproductive success.

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